Random Variables

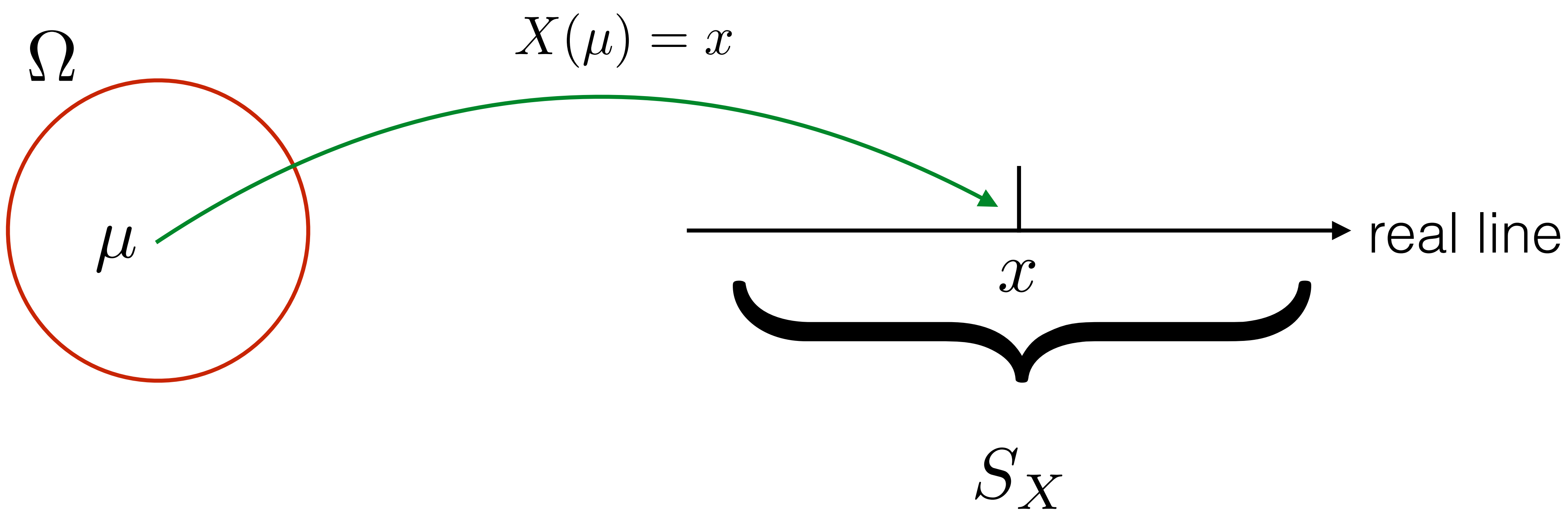
Lecture 4

Topics in this lecture

- Notion of random variables
- Discrete random variables

Random Variable

- Frequently, we need to assign a numerical value to the outcome of the experiment
- A random variable X is a *function* that assigns a real number, $X(\mu)$ to each outcome μ in the sample space of a random experiment



Example 4.1

- A coin is tossed 3 times and a sequence of heads and tails is noted. Let X be the number of heads in 3 coin tosses. Find the sample space Ω of the experiments, and set S_X of all values taken by X.

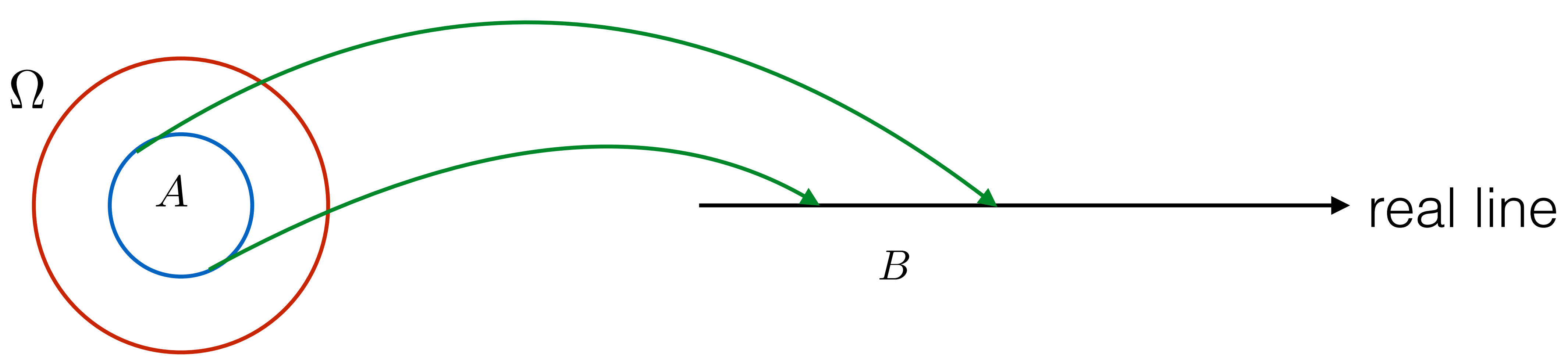
$S_X = \{0, 1, 2, 3\}$

Probability of Events in S_X

- Let S_X be the set of values that can be taken on by X. Let B be some subset of S_X. S_X can be viewed as a new sample space, and B as an event in the sample space.

- Let A be the set of outcomes μ in Ω that lead to values $X(\mu)$ in B, then the event B in S_X occurs whenever the event A in Ω occurs.

- A and B are equivalent events, and $P(A) = P(B)$



Computing Probability Relating to a Random Variable

- We use a *distribution* to describe a random variable
- The distribution functions for discrete random variable
    - Probability Mass Function (PMF)
    - Cumulative Distribution Function (CDF)

Probability Mass Function

- For *discrete* random variable only
- Probability mass function (PMF) of X is the set of probabilities $p_X(x) = P(\{X = x\})$ of the elements in S_X
- The probability mass of x, denoted by $p_X(x)$, is the probability of the event $P(\{X = x\})$ consisting of all outcomes that give rise to a value of X equal to x

Important Note

- The following condition needs to be true for a valid PMF

$$\sum_{x} p_X(x) = 1$$

where *x* ranges over all the possible numerical values of *X*

Example 4.2

- Toss a fair coin twice. Let X be the number of heads in two tosses.
    - Find the PMF of X
    - Find the probability that there is at least one head in two tosses

Example 4.3

- Roll a fair die twice. Let X be the sum obtained from the two rolls.
    - Find the PMF of X
    - Find the probability that the sum obtained from the two rolls is less than 5

Example 4.4

- A packet is transmitted from a source to a destination. The source will keep transmitting the packet until it is received correctly. Let *X* be the number of times required to transmit a packet. Assuming that the probability that a packet will be received correctly is 0.2.

$S_X = \{1, 2, 3, \dots\}$

$P(X=1) = 0.2$

$P(X=2) = 0.8 \times 0.2$

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$P(X=3) = (0.8)^2 \times 0.2$

  - Find the PMF of *X*

  - Find the probability that the packet will need to be transmitted more than 4 times

$$P_X(x) = \begin{cases} 0.8^{x-1} \times 0.2 & ; x \in \{1, 2, 3, \dots\} \\ 0 & ; \text{ฯลฯ} \end{cases}$$

$P(X>4) = 1 - P(X \leq 4)$

$P(X>4) = 1 - (0.8^3 \times 0.2 \times 0.8^2 \times 0.2 \times 0.8 \times 0.2 \times 0.2)$

Example 4.5

- An urn contains 2 red balls and 3 white balls. Select 3 balls from an urn with replacement. Let X be the number of white balls selected.
  - Find the PMF of X
  - Find the probability that none of the white balls are selected

Important Discrete Random Variables

- Bernoulli Random Variable

$$S_X = \{0, 1\}$$

$$P(X = 0) = 1 - p$$
$$P(X = 1) = p$$

- Binomial Random Variable

$$S_X = \{0, 1, \ldots, n\}$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Important Discrete Random Variables (2)

- Geometric Random Variable

$$S_X = \{1, 2, \ldots\}$$

$$P(X = k) = p(1-p)^{k-1}$$

- Poisson Random Variable

$$S_X = \{0, 1, 2, \ldots\}$$

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

Functions of RV

- Given a random variable X, one may generate other random variables by applying various transformation on X
- Define $Y = g(X)$, that is, Y is determined by evaluating the function $g(x)$ at the value assumed by the random variable X. Then, Y is also a random variable.
- If X is discrete with PMF $p_X(x)$, then Y is also discrete and its PMF $p_Y(y)$, can be calculated using $p_X(x)$

$$p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$$

Example 4.6

- Let X be a random variable with the following PMF.

$$p_X(x) = \begin{cases} 1/7 & \text{, if } x \text{ is an integer in [-3,3]} \\ 0 & \text{, otherwise} \end{cases}$$

$$p_Y(y) = \begin{cases} \frac{1}{7} & = y = 0 \\ \frac{2}{7} & = y = 1, 2, 3 \\ 0 & \text{o.w.} \end{cases}$$

Let $Y = |X|$, find PMF of Y.

$S_Y = \{0, 1, 2, 3\}$

$P(Y=0) = P(X=0) = \frac{1}{7}$

$P(Y=1) = P(X=1) + P(X=-1) = \frac{2}{7}$

$P(Y=2), P(Y=3) = \frac{2}{7}, \frac{2}{7}$

$\frac{7}{7} = 1$

Example 4.7

- Let X be a random variable with the following PMF.

$$p_X(x) = \begin{cases} 1/7 & \text{, if } x \text{ is an integer in [-3,3]} \\ 0 & \text{, otherwise} \end{cases}$$

Let $Z = X^2$, find PMF of Z.

Expected Value

- In some situations, we are interested in a few parameters that summarize the information provided by PMF

- The expected value or the mean of a random variable X with PMF $p_X(x)$ is defined as

$$\boxed{E[X] = \sum_x x p_X(x)}$$

Example 4.8

1. หา pmf

- Let X be the number of heads obtained from 2 independent coin tosses. If the probability of getting head in each toss is 2/3. Find the expected value of X.

$S_X = \{0, 1, 2\}$

$P(X=0) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$

$P(X=1) = \frac{2}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3} = \frac{4}{9}$

$P(X=2) = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

$$p_X(x) = \begin{cases} \frac{1}{9} & ; x = 0 \\ \frac{4}{9} & ; x = 1, 2 \\ 0 & \text{อื่นๆ} \end{cases}$$

$+ \frac{8}{9} + 4$

Example 4.9

- Roll two dice and observe the sum of the numbers on the dice. Find the expected value of the sum.

Expected Value of $Y = g(X)$

- The expected value of Y can be obtained from PMF of X as follows

$$E[Y] = \sum_x g(x) p_X(x)$$

Example 4.10

- Let X be the number of heads obtained from 2 independent coin tosses. If the probability of getting head in each toss is 2/3 and $Y = 2X + 3$. Find the expected value of Y.

Properties of Expected Value

- $E[c] = c \quad ; \quad c \text{ is a constant}$

- $E[cX] = cE[X] \quad ; \quad c \text{ is a constant}$

- $E[X + c] = E[X] + c \quad ; \quad c \text{ is a constant}$

Variance of *X*

- The variance of the random variable *X* is defined as

$$\mathrm{VAR}[X] = \sigma_X^2 = E[(X - E[X])^2]$$

- The standard deviation of the random variable is defined as

$$\mathrm{STD}[X] = \sigma_X = \sqrt{\mathrm{VAR}[X]}$$

Example 4.11

- Let X be the number of heads obtained from 2 independent coin tosses. If the probability of getting head in each toss is 2/3. Find the variance of X.

Properties of Variance

- $\mathrm{VAR}[c] = 0 \quad ; \quad c$ is a constant
- $\mathrm{VAR}[cX] = c^2\mathrm{VAR}[X] \quad ; \quad c$ is a constant
- $\mathrm{VAR}[X + c] = \mathrm{VAR}[X] \quad ; \quad c$ is a constant